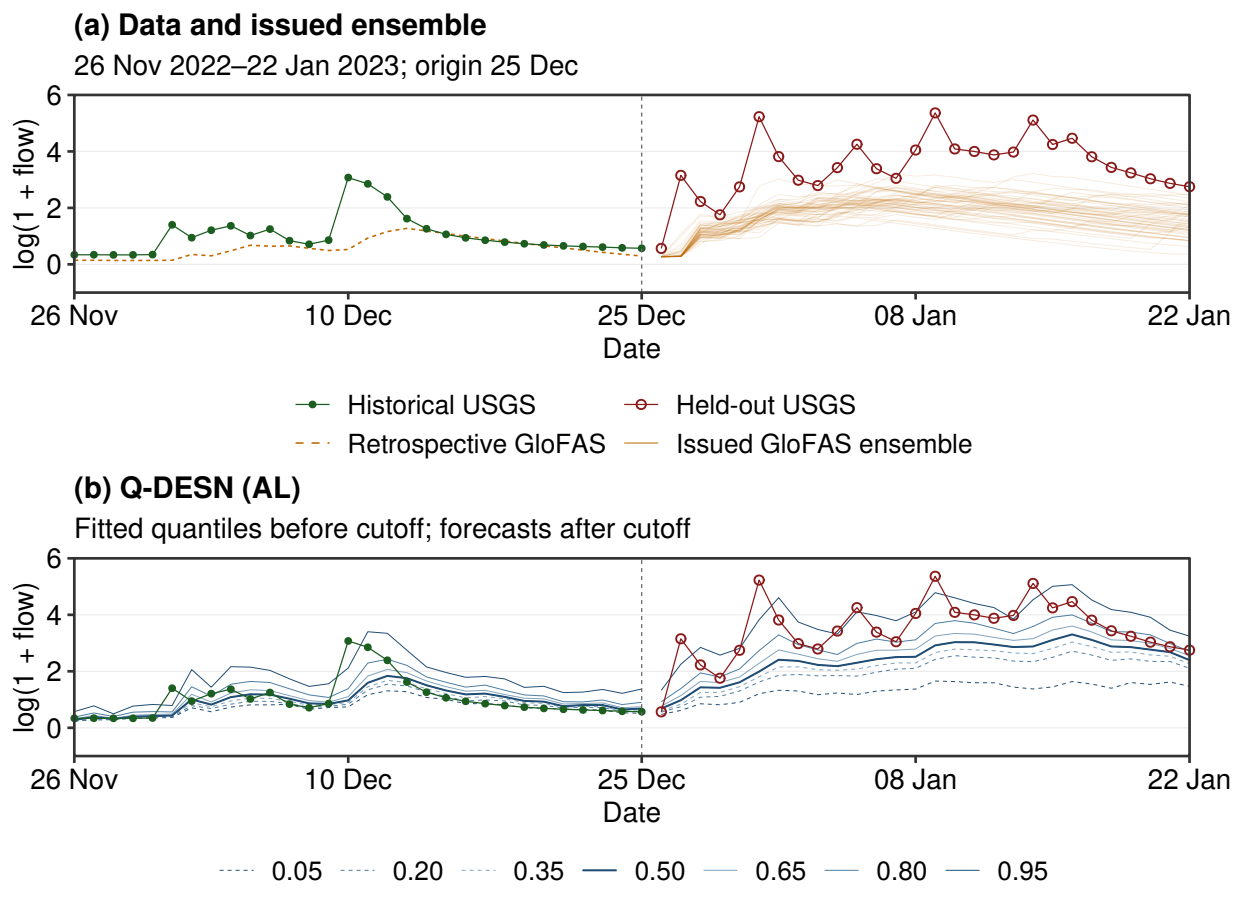

(a) Data and issued ensemble
26 Nov 2022–22 Jan 2023; origin 25 Dec
log(1 + flow)
6
4
2
0
26 Nov
10 Dec
25 Dec
08 Jan
22 Jan
Date
Historical USGS
Held-out USGS
Retrospective GloFAS
Issued GloFAS ensemble
(b) Q-DESN (AL)
Fitted quantiles before cutoff; forecasts after cutoff
log(1 + flow)
6
4
2
0
26 Nov
10 Dec
25 Dec
08 Jan
22 Jan
Date
0.05
0.20
0.35
0.50
0.65
0.80
0.95